



Supply Chain Operations Reference Model for the Flight Simulator Maintenance Service Process Model: A Case Study of Asian Aviation Training Centre

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ABSTRACT

This study aims to present the flight simulator maintenance process model. To avoid failures of the system and mechanical, it is necessary to implement the maintenance processes. Moreover, flight simulator operator has to ensure that the performance of the devices qualified with the international regulations published by EASA, ICAO, and FAA. This paper demonstrates the maintenance process model and framework which is adapted to the supply chain operations reference (SCOR) model, to implement the flight simulator maintenance supply chain. The maintenance process model purposed with the most popular notation which called Business Process Model and Notation (BPMN) to support identification and visualisation of the processes as well as validate the effectiveness of models through the reviewing and analysing methodology with the experts in the field.

CCS Concepts

• Social and professional topics → Systems planning • Social and professional topics → Systems development

Keywords

Service supply chain; Maintenance; Flight simulator maintenance; SCOR model; BPMN

1. INTRODUCTION

Business aviation well-established in the 1920s, it is a part of general aviation focusing on the business use of airplanes and helicopters. The business aviation has been growing to be a substantial contributor because of the high level of its benefits. Flying is an exclusive experience; it becomes the alternative primary traveling. Allowing across vast distances with low-priced made the number of passengers handled by the airline industry globally from 2004 to 2018 is growing up. The growth of passenger and air cargo traffic rates from 2018-2037 are expected to grow by 4.7 percent [1] as shown in Figure 1. According to

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lower oil prices and the result of increasingly affordable flight rates, and growing up of low-cost airline, passenger and cargo air traffic, the growth passenger is estimated to grow importantly through 2037.

Additionally, Thailand's aviation business growth is also increasing; it is rising three times faster than the global market. The number of visitors to Thailand increase as well as the travelers in domestic routes, so the expansion of Thai Airlines is mainly driven by the demand for low-cost carriers. Thailand Board of Investment 2016 claimed that the market share of low-cost airlines currently contributes to only about 30% of the total market share, which showcases the vast growth opportunity in this sector.

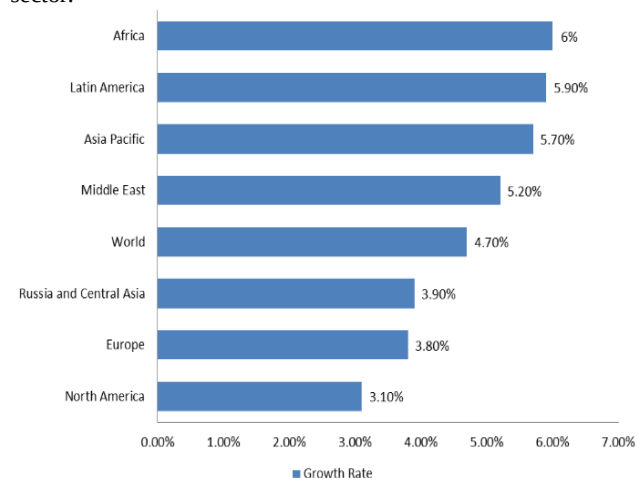


Figure 1. Annual growth rates for passenger air traffic from 2018 to 2037.

Growing of business of aviation is not just only the increasing of airlines revenue, but it included all components of aviation industry which growth with aviation's company such as aircraft manufacturer, energy, insurance, technology, and human resources. Human resources represent another critical consideration concerning growing aviation's business. In such a safety requirement from aviation standard, human resources expertise such as pilots, engineering maintenances, cabin crews, ground services, and air traffic controllers required for the aviation industry. Aviation industry jobs are specifically job which needs to attend competency training; thus, flight training academy is one of the businesses that has been growing in the aviation industry. At the moment, pilot and mechanics are the most demand in airline jobs. Right now, careers in aviation

account for over \$1.5 trillion in the U.S economy [2], there was no doubt about being new careers with new skills requirements would earn more pay raise.

Aircraft crashes are the most disastrous of all transport accidents, primarily because they often involve a catastrophic loss of life. Aircraft crashing can kill as many as four or five hundred times more people than a car accident. Therefore, global aviation organisations emphasise the standard of aviation training to meet aviation priorities and implement the specific instruction to develop human resources to achieve aviation safety's standard.

2. LITERATURE REVIEWS

2.1 Aviation Industry

It was over the past 50 year that the aviation industry had grown exponentially due to escalating global travel demand. The revenue of the civil aviation flight training market is expected to reach 8.3 billion dollars by 2023 [3]. Increasing demand for air travel and the needed for the pilot have significantly improved, observed from the statistic of demand for commercial airline pilots in 2016-2035 especially in Asia Pacific region; it has increased 40% of world total [4]. This qualified pilot demand also increased more and more the need for the flight simulation market.

Becoming a pilot in Thailand, a bachelor's degree in aircraft operations, aviation, aeronautical engineering, or a related field is required. The skills needed for this career include secure communication, problem-solving and observation skills, good depth perception and reaction time, and the ability to operate aircraft computer and navigation systems. The key to becoming a good pilot is getting a certified pilot license before operating an aircraft; therefore, global aviation needs an academy for developing qualified pilots. It is a very complicated process to be a pilot due to safety regulations as illustrated in Figure 2.1.



Figure 2.1 Pilot Supply Chain

Aspiring airline pilots and private pilots are required to complete a certain number of hours of flight training simulator to qualify for licensure. To obtain a commercial pilot's license, Civil Aviation Authorities of Thailand (CAAT) requires 32 hours of flight simulator training time for initial training, once became a pilot, the pilots are generally required to recurrent pilot's license six-monthly, by training in a flight simulator for 4 hours.

The flight training simulation is simulated real flying by using motion and visual systems; the majority of flight training devices are used for pilot training. The use of flight simulators is widely accepted around the world because one of the essential benefits of flight simulator training is training time in a simulator can replace training time in an aircraft. And these are the significant factors that drove the revenue of training simulation market include rising fuel prices and the growing international regulation of emissions and noise pollution.

The development of flight simulator began since World War I; it was developed and evolved to further technology continuously. The first known flight simulation device was the Antoinette monoplane, which helped the pilots to simulate the flight [5]. In 1927, Edwin Link built the flight simulation called the Link Trainer which provided a pneumatic motion platform driven by inflatable bellows. The Link Trainer gave the pilot cues as to

original angular motion in pitch, roll, and yaw, consequently, the U.S. Army Air Corps purchased six Link Trainers [5], this is the start of the world flight simulation industry, and it finally was developed to a full flight simulator (FFS) nowadays.

Asian Aviation Training Centre (AATC) is Asia's leading independent Approved Training Organisation (EASA.ATO.0024) [6]. AATC provides approved training for pilots combining state-of-the-art Full Flight Simulation (see Figure 2.2) technology with enhanced Flat Panel Trainer technology and operates twenty-four hours a day, seven days a week. To provide quality service, to achieve the highest customer satisfaction and to prepare for the increase of flight training market, AATC would like to develop a new maintenance process, which is an integrated system for flight training reservation, maintenance management, shipping and store management. Operation and installation all the flight simulators must be provided under the current regulatory standards, criteria, and requirement of aviation regulation from ICAO and EASA.



Figure 2.2 a Full Flight Simulator

Although, the flight simulator training devices are always fully booked but each flight training device needs to be maintained and monitored regularly, especially, working twenty-four hours and seven days of the devices. Any technical issues of a flight simulator cause not only trouble to engineers but can also lead to serious problems for training centers. Therefore, engineers usually perform Daily, Weekly, Monthly, Quarterly, Six Monthly and Yearly routine maintenance of simulators to ensure uninterrupted operation and reduce the risk of any failures during pilot training [7].

In case of maintenance services, the item flowing toward the customer will be serviced, not physical products as illustrated in Figure 2.3. The concept of introducing the customer in the actual chain should be where the customer takes an active part in adding value to the service [8].

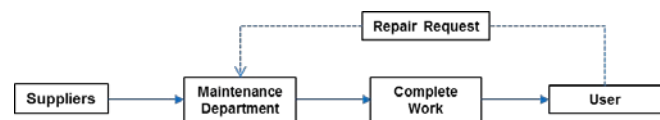
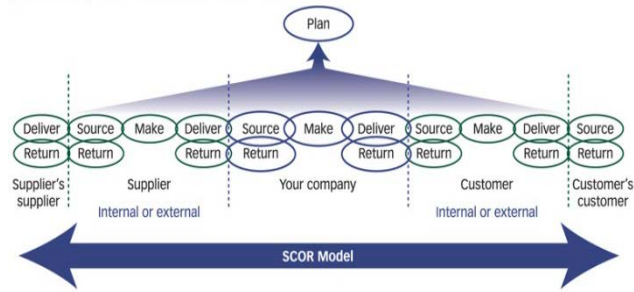


Figure 2.3 Maintenance Service Supply Chain

2.2 SCOR Model

The SCOR modeling framework is the standard implementation roadmap which has been created and developed by the Supply Chain Council (SCC) for the supply chain standardisation, measurement, and improvement. The SCOR model combines four essential elements including Process, Executions, Metrics, and

Best practice. The structure of the SCOR model is classified to



five basic processes, called: Plan (P), Source (S), Make (M), Deliver (D) and Return (R) as illustrated in Figure 2.4.

Figure 2.4 SCOR Model

The SCOR model mainly contains three levels of process details. Level 1 is the top level that deals with process types which define a scope of a supply chain. Level 2 is the configuration level which deals with process categories. The lowest level in the SCOR model is Level 3 which is the processing element or executes.

In SCOR, each process has its performance metrics to monitor overall supply chain performance which attributes: Reliability, Responsiveness, Agility, Cost, and Assets management efficiency. Each process is measured against these performance attributes; it has been widely used in the global organisation and commonly accepted.

Although the SCOR model was designed and applied for manufacturing industries, based on identified limits and weaknesses, researches and results from different practitioners and academicians such as Weyers and Ellram proved that SCOR model could be extended and increasing its proven practical relevance. Therefore, Weyers' approach model was adapted by using semantics, removal of elements and limited addition of elements that relate to services. The study claimed that the SCOR model could be created for services business and the model was useful to participants that work with back office services.

2.3 Business Process Modeling Notation

BPMN stands for Business Process Modeling Notation which is a graphical representation for specifying business processes. The purpose of BPMN is to facilitate communication and understanding business processes, and the BPMN is a language which can be easy to understand for the business industry. It is an effective and useful methodology to use in crisis time because the BPM can help an organisation to look at and control the processes that represented in the organisation, it will show a better and more cost efficient organisation. The simplest way of business process management is workflow which is known as routing. Workflow is more than telling moving things from process A to process B to process C to process D or combine all tasks because it also carries out functions in parallel, increase productivity and save time. It is reasonable to say that BPMN is one of the most popular business process languages.

2.4 Semantics Changed

The SCOR model adaptation will be adapting the model through the changing of words, using a direct translation or similar concepts. An example of the semantics changed is the definition and use of the "Make" process. Semantically, the "Make" definition in SCOR is the process where the value added to the

product. Translating this "Make" concept to the service sector creates a situation where some of the intent and concepts, specific to manufacturing, may be lost in the translation to a new concept. Thus, there is no translation of "Make" relate in the service business.

In this procedure will analyse the original SCOR process name each elements which comprised of Plan, Source, Make, Deliver, and Return through Weyers' approach, each descriptions examined with the decision of the adaptation approach options, any process elements not relevant with maintenance service process will be removed or changed the title of the processing element. An example of adapted maintenance service process elements through SCOR model is derived with the new process elements illustrated in Table 1.1 and Table 1.2 respectively.

2.5 Adapted SCOR Model

The previous section changed the name of elements processes, based on the above changes; the new process references are msP (maintenance process Plan), msS (maintenance service Source), msF (maintenance service Fulfil), msD (maintenance service Deliver), and msR (maintenance service Return) respectively. Besides the new process elements, the maintenance services classified into Scheduled and Unscheduled follow preventive and corrective maintenance; thus Figure 2.5 summarises the adapted maintenance service SCOR model.

SCOR Processes						
		Plan	Source	Fulfill	Deliver	Return
SCOR Processes	Plan	msP1 Plan Supply Chain	msP2 Plan Source	msP3 Plan Fulfil	msP4 Plan Deliver	msP5 Plan Return
	Execute		msS1 Source Stock Spare part	msF1 Fulfil Schedule Service	msD1 Deliver Schedule Service	msRE1 Request Return Service
			msS2 Source Ordering Spare Part	msF2 Fulfil Unschedule Service	msD2 Deliver Unschedule Service	msRE2 Deliver Return Service
			msS3 Source Engineered Service	msF3 Fulfil Engineered Service	msD3 Deliver Engineered Service	msRE3 Deliver Return MRO Product

Figure 2.5 Adapted Service SCOR Process Elements

After the adapted process elements in the scope of the SCOR model have been generated. The next step will elaborate on the SCOR Level 1 and 2; the adapting methods in the process of maintenance service supply chain have followed the design of the SCOR model in total five process elements (Plan, Source, Fulfil, Deliver, and Return) as shown in Figure 2.6.

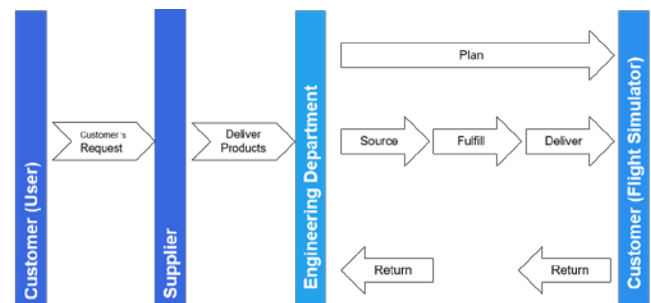


Figure 2.6 an Overview Adapted SCOR Model Process Elements

Table 1.1 Plan Process Approach

Process Reference	Original Process Name	Process Removed	New Process Reference	New Process Name	No Title Change	Semantic
sP1	Plan Supply Chain		msP1	Plan Maintenance Service Supply Chain		x
sp1.1	Identify, Prioritize and Aggregate Supply Chain Require...		msP1.1	Identify, Prioritize and Aggregate MS Supply Chain...	x	
sp1.2	Identify, Prioritize and Aggregate Supply Chain Resources		msP1.2	Identify, Prioritize and Aggregate MS Supply Chain...	x	
sp1.3	Balance Supply Chain Resources with Supply Chain...		msP1.3	Balance MS Supply Chain Resources with MS Supply...		x
sp1.4	Establish and Communicate Supply Chain Plans		msP1.4	Establish and Communicate MS Supply Chain Plans		x
sP2	Plan Source		msP2	Plan Source	x	
sp2.1	Identify, Prioritize and Aggregate Product Requirement		msP2.1	Identify, Prioritize and Aggregate Source Services...		x
sp2.2	Identify, Assess and Aggregate Product Resources		msP2.2	Identify, Assess and Aggregate Services Resources		x
sp2.3	Balance Product Resources with Product Requirements		msP2.3	Balance Service Resources with Service Requirements		x
sp2.4	Establish Sourcing Plans		msP2.4	Establish and Communicate Sourcing Plans		x
sP3	Plan Make		msP3	Plan Fulfil	x	
sp3.1	Identify, Prioritize and Aggregate Production Requirement		msP3.1	Identify, Prioritize and Aggregate Fulfil Requirement		x
sp3.2	Identify, Assess and Aggregate Production Resources		msP3.2	Identify, Assess and Aggregate Fulfil Resources		x
sp3.3	Balance Production Resources with Production...		msP3.3	Balance Fulfilment Resources with Fulfilment...		x
sp3.4	Establish Production Plans		msP3.4	Establish and Communicate Fulfil Plans		x
sP4	Plan Deliver		msP4	Plan Deliver	x	
sp4.1	Identify, Prioritize and Aggregate Delivery Requirement		msP4.1	Identify, Prioritize and Aggregate Delivery Requirement	x	
sp4.2	Identify, Assess and Aggregate Delivery Resources		msP4.2	Identify, Assess and Aggregate Delivery Resources	x	
sp4.3	Balance Delivery Resources and Capabilities...		msP4.3	Balance Delivery Resources and Capabilities...	x	
sp4.4	Establish Delivery Plans		msP4.4	Establish and Communicate Delivery Plans		x
sP5	Plan Return		msP5	Plan Return		
sp5.1	Assess and Aggregate Return Requirements		msP5.1	Assess and Aggregate Return Requirements	x	
sp5.2	Identify, Assess and Aggregate Return Resources		msP5.2	Identify, Assess and Aggregate Return Resources	x	
sp5.3	Balance Return Resources with Return Requirement		msP5.3	Balance Return Resources with Return Requirement	x	
sp5.4	Establish and Communicate Return Plans		msP5.4	Establish and Communicate Return Plans	x	

Table 1.2 Return Process Approach

Process Reference	Original Process Name	Process Removed	New Process Reference	New Process Name	No Title Change	Semantic
sSR1	Source Return Defective Product		msR1	Source Return Defective Service		x
sSR1.1	Identify Defective Product Condition		msR1.1	Identify Defective Service Condition		x
sSR1.2	Disposition Defective Product		msR1.2	Disposition Defective Service		x
sSR1.3	Request Defective Product Return Authorization	x				
sSR1.4	Schedule Defective Product Shipment	x				
sSR1.5	Return Defective Product		msR1.3	Return Defective Service		x
sDR1	Deliver Return Defective Product		msR2	Deliver Return Service		x
sDR1.1	Authorize Defective Product Return		msR2.1	Authorize Defective Service Return		x
sDR1.2	Schedule Defective Return Receipt					
sDR1.3	Receive Defective Product (includes verify)		msR2.2	Receive Defective Service (includes verify)		x
sDR1.4	Transfer Defective Product		msR2.3	Transfer Defective Service		x
sSR2	Return MRO Product	x				
sDR2	Deliver Return MRO Product					
sSR3	Source Return Excess Product	x				
sDR3	Deliver Return Excess Product	x				

3. APPLICATION SCOR IN FLIGHT SIMULATOR MAINTENANCE SERVICE

As the AATC never organised the maintenance service model before, the maintenance workflows both of preventive and corrective maintenance has conducted by interviewed the Engineering management teams and Supervisor Engineers, then analyse maintenance service workflow to maintenance service

process model. The maintenance workflow was generated from the discussion and interviews the staffs in the company as summaries in Figure 3.1, the workflow divided into two categories: Corrective maintenance service and Preventive maintenance service. The maintenance process “As-Is” analysis started with the original maintenance workflow then the model would be adapted to SCOR model level 1, level 2, and level3 respectively.

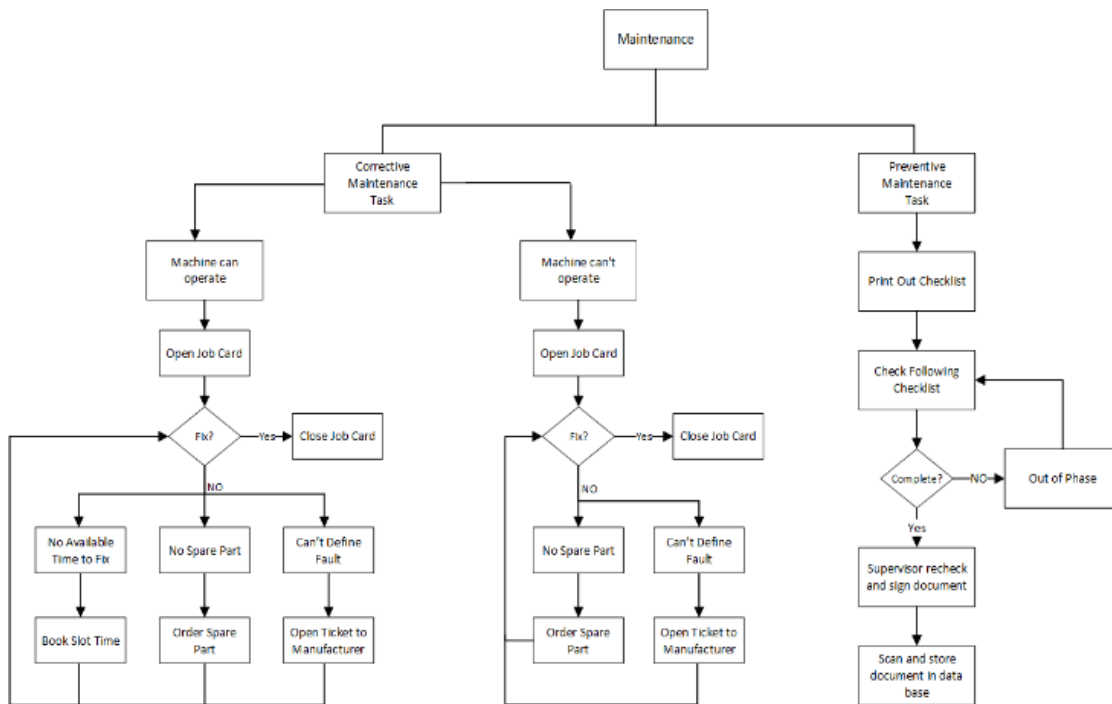


Figure 3.1 Maintenance Workflow

Corrective maintenance generally starts from receiving a call from the customer to report faults, then categories into two cases: the FFS can operate and cannot operate. After that the engineer will discuss and inform a problem to customers and sourcing a spare part as well as an application engineer for preparing. And then the engineer will go to solve the problem or in some cases, the suppliers will in charge of the problem.

On the other hand, preventive maintenance starts from print maintenance checklist out, then checking to follow the activities on the list. The preventive maintenance activities should be finished on time follow the cycle such as weekly, monthly, or even annually. The significant factors which effect on preventive maintenance activities are a weak schedule, spare part shortage, and uncontrollable issues such as an electric power system down. The engineering supervisor will in charge of out of phase tasks.

The SCOR Level 1 as shown in Figure 3.2 corresponds the overall of plan processes which concern with management or top level. The process described the maintenance service flow and information flow between the engineer and suppliers and the engineer and user.

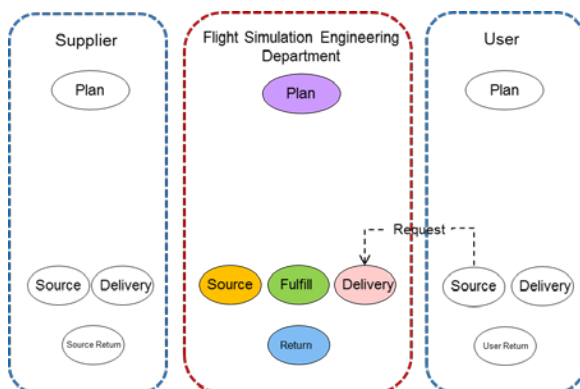


Figure 3.2 the SCOR Level 1 Maintenance Service Process

The SCOR Level 2 break down from level 1, the process in level 2 integrate all participates and activities together. The integration along the maintenance service supply chain generated both information flow and physical flow is shown in Figure 3.3. The level 2 process classified in 5 processes:

- Plan (msP): the process relevant to the integrated maintenance service supplies chain.
- Source (msS): the process is verifying suppliers, spare parts, and user's requirements
- Fulfil (msF): the process adds value to fulfill user requirements before transferring the tasks to Deliver.
- Deliver (msD): the process that consists of fixing, testing and delivering fulfilled services to the user.
- Return (msR): the process to determine the defective sources and the services that do not meet the requirements.

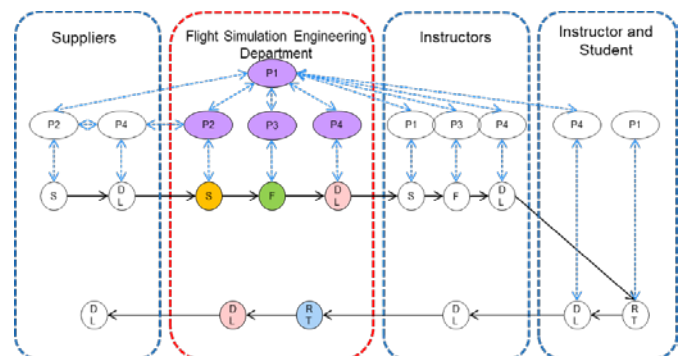


Figure 3.3 the SCOR Level 2 Maintenance Service Process

The SCOR Level 3 applied through BPMN tools has shown the preventive maintenance and corrective maintenance service process model as illustrated in Figure 3.4 and 3.5 respectively. The SCOR level 3 model break down processes from level 2 to the maintenance service execution level to define the individual

processes. The corrective maintenance service process is a complicated model due to connecting with various suppliers and internal departments, to design the unscheduled maintenance service process need to analyse the task in each activity and sections.

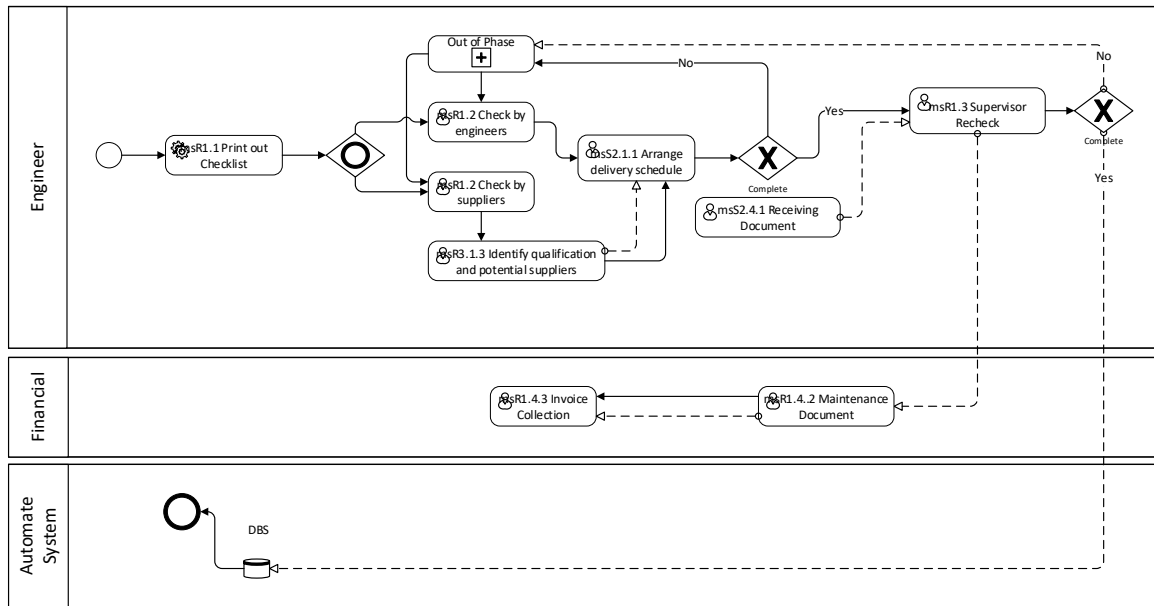


Figure 3.4 the SCOR Level 3 of Preventive Maintenance Service

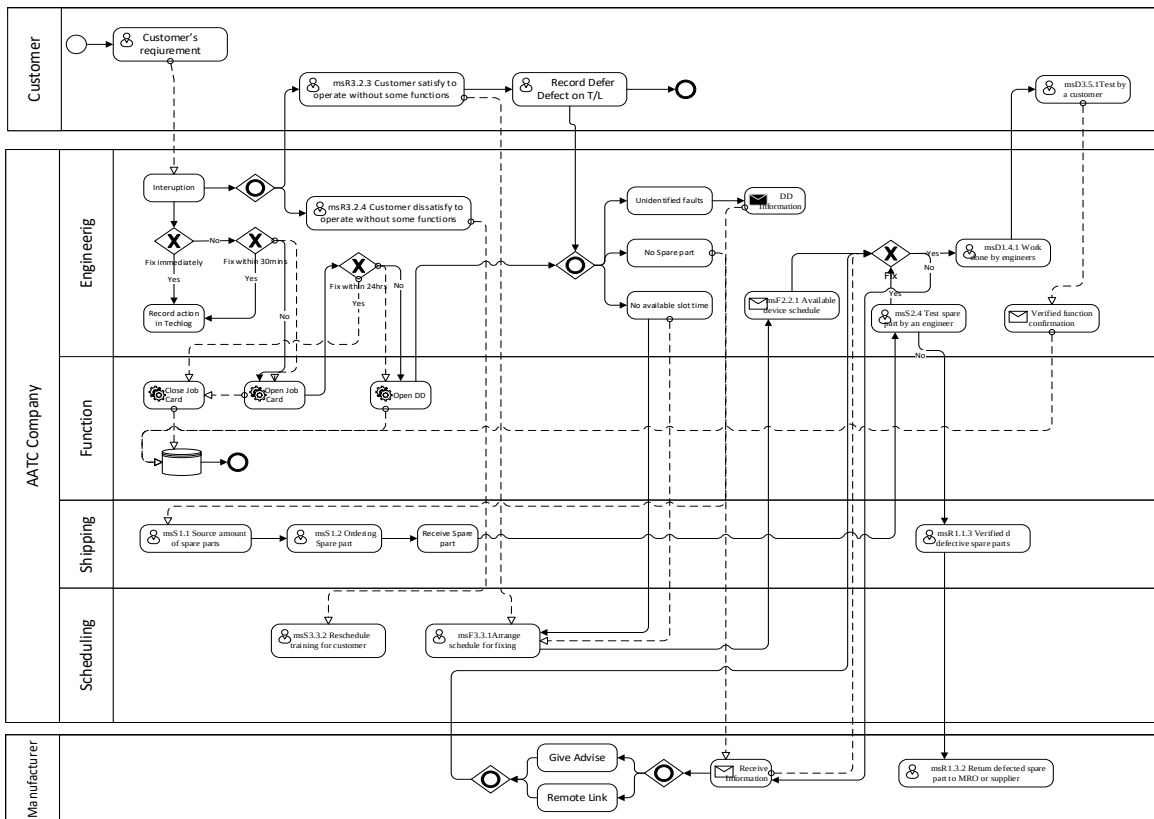


Figure 3.5 the SCOR Level 3 of Corrective Maintenance Servi

The model validation and testing base on the review from 20 expertise in a case study company through an evaluation methodology for the business process [11] designed by S. Adesola and T. Baines. Testing methodology was refined to applicable with maintenance service model testing then the questionnaire was set out to determine the following [11]:

- Could the methodology be used in practice?
- Are there any problems and difficulties with the model?
- Are the designed models worth the effort and are they useful to the host organisation?

To assess whether the methodology was working, a procedural approach of evaluating process research was adopted Platts (1990) [12]. The subsequent assessment procedure consisted of three categories of measurement and were [11]:

- Feasibility: Can the adapted model be followed?
- Usability: Is the adapted model workable? Are the steps and techniques easy to use and apply?
- Usefulness: Is the adopted model worth following? Does the model produce results that the business finds helpful?

Evaluation results base on the description above of feasibility, usability, and usefulness, the conclusions were shown that the company would benefit from being developed designed model. In addition, the engineering team can be a part of designing their procedures base on the SCOR model. Maintenance service process improvement was developed through the feedback from expertise so that the maintenance service model processes earn more effectively and efficiently.

4. CONCLUSIONS

In conclusion, the designing of the adapted maintenance service process model can create a standardised maintenance services process model base on the SCOR model. The study has reached its objectives and shown that the model can be improved service level and reduced the breakdown. Moreover, the models through BPMN can be useful for business management level for implementation and communication between levels and suppliers.

In this study, a major limitation is a time-limited, thus, the models were applied only on the management level. As well as the case study is a specific unit; this may mean the maintenance service process model does not apply to general businesses. An important suggestion for future work is to gather more data and take more time to design the performance measurement system. This could increase the reliability of the maintenance service.

5. ACKNOWLEDGMENTS

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